



MUSCULOSKELETAL INFECTIONS- PART B MUSCLE INFECTIONS

CHITRA PAI, MD, D(ABMM)
Professor of Microbiology
TUCOM-CA



LEARNING AND TESTABLE OBJECTIVES

- List the major pathogens of myositis
 - *Streptococcus pyogenes*
 - *Clostridium botulinum*
 - *Clostridium tetani*
 - *Trichinella spiralis*
 - *Cysticercus (Taenia spp.)*
- Discuss pertinent characteristics of the pathogens including,
 - Morphological and cultural characteristics
 - Source, transmission, and relevant epidemiology
 - Virulence factors and pathogenesis
 - Disease association and complications
 - Laboratory Diagnosis
 - Treatment, Prevention

INFECTIOUS MYOSITIS BUG LIST

BACTERIA	FUNGI	PARASITES
<i>Staphylococcus aureus</i>	<i>Candida spp</i> ★	<i>Trichinella spiralis</i> ★
<i>Streptococcus pyogenes</i> ★	Others: <i>Cryptococcus spp</i>	Cysticerci of <i>Taenia spp</i> (tapeworm larvae) ★
<i>Clostridium botulinum</i> ★		Others: <i>Trypanosoma cruzi</i> (causes Chaga's disease); <i>Toxoplasma gondii</i>
<i>Clostridium tetani</i> ★		
Others- <i>Borrelia burgdorferi</i> <i>Mycobacteria</i> , other non- sporing anaerobes		

cardiac muscle

focus on red stars

BACTERIAL INFECTIONS

Pyomyositis

- Purulent condition of the skeletal muscles.
- Localized to single muscle group.
- Polymicrobial/ Monomicrobial
- **#1 pathogen:** *Staphylococcus aureus*

Necrotizing fasciitis

- Rare but rapidly progressive and fatal.
- Involves extremities, abdominal wall or perineum
- Poly/ Mono microbial
- **Pathogen:** *Streptococcus pyogenes* (GAS- Group A Strep) causes majority cases.

Necrotizing myositis

- **GAS GANGRENE** or clostridial myonecrosis
- Uncommon but rapidly progressive and life threatening
- Polymicrobial
 - Several species of Clostridia: *C. perfringens*, *C. septicum*, *C. novyi*, etc. along with GPC and GNB.

Tetanus

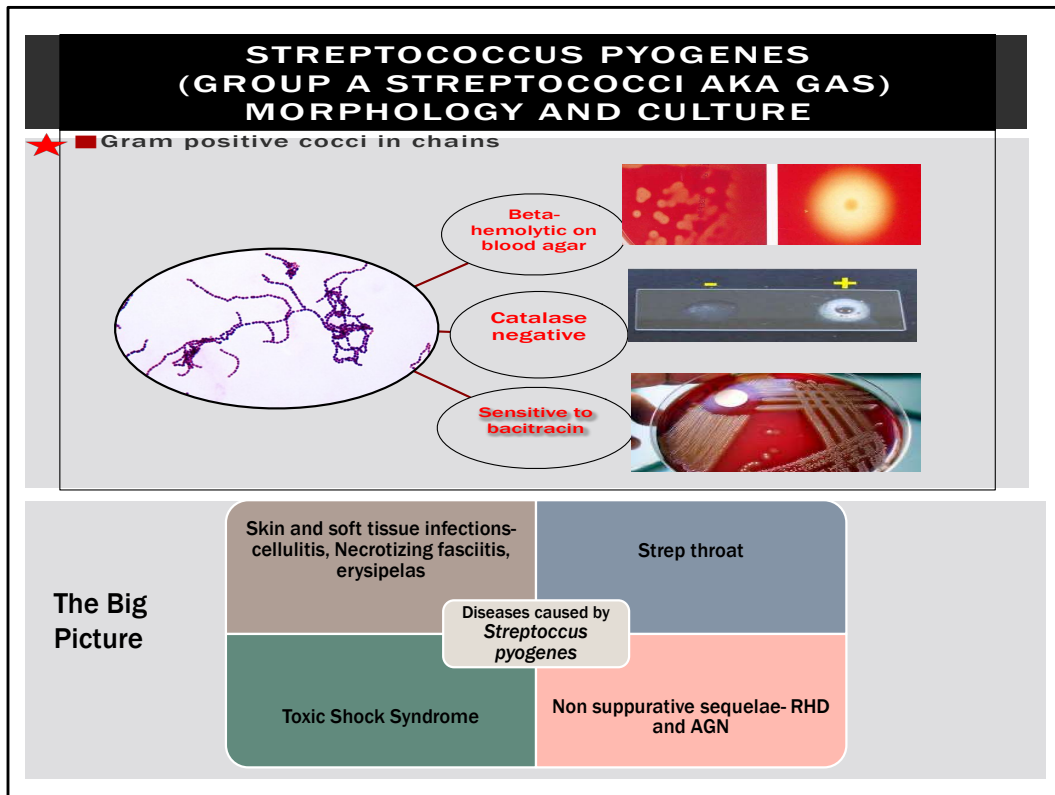
- Neuromuscular condition
- **Pathogen-** *Clostridium tetani*
- Spastic paralysis due to toxins
- can be fatal

NECROSTIZING FASCITIS

- A rare bacterial infection that spreads quickly.
- Associated with cuts/trauma/burns/ patients with diabetes
- Poly/ Monomicrobial. Most important pathogens:
 - Group A Streptococci (GAS)- *S.pyogenes* (*referred to as flesh eating bacteria*)
 - Can also involve MRSA, Enterococci, GNB etc.



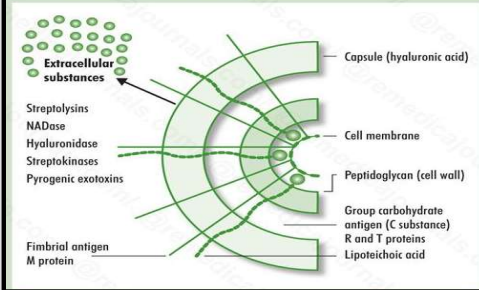
Strep
rheumatic heart
disease,
TSS



strep group B (agalactiae) is resistant to bacitracin and causes skin infec

VIRULENCE FACTORS PATHOGENESIS

Figure 1. Cell surface structure of *Streptococcus pyogenes* and secreted products that are involved in virulence.



Entry of Strep via wound

Adherence to epithelial cells at site of wound

LTA-M protein complex bind to Fibronectin

Multiplication while evading phagocytosis

M proteins esp. M1 and M3 types

Rapid spread due to hyaluronidase and hydrolytic enzymes

Further damage due to toxins like Streptolysins

Possible **Complication-Toxic Shock Syndrome**
(Vir Fac- Streptococcal Pyrogenic Exotoxin (TSST))

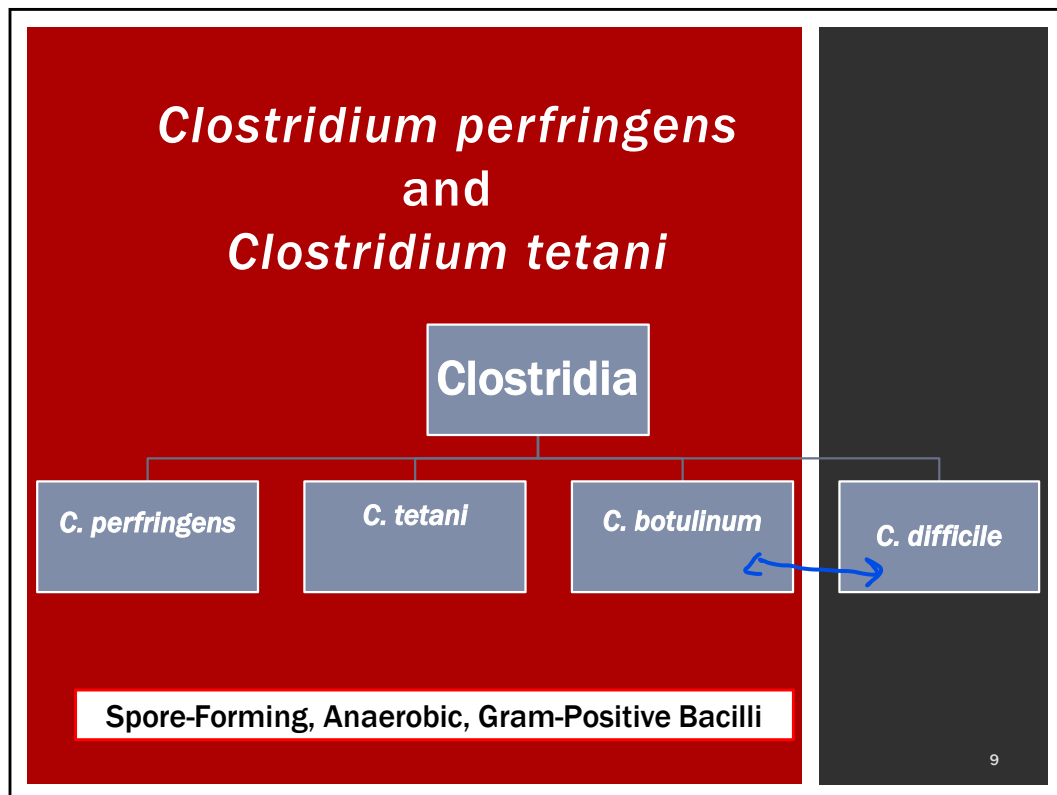
CLINICAL FEATURES DIAGNOSIS, TREATMENT

C/F

- **Fulminant tissue destruction, systemic signs of toxicity, and high mortality.**
- **Fever, extensive and rapidly progressive redness, pain, swelling, bullae and necrosis of area involved-usually limbs.**
- **Complications- TSS/sepsis and organ failure**
- **Important differential diagnosis for Gas gangrene**

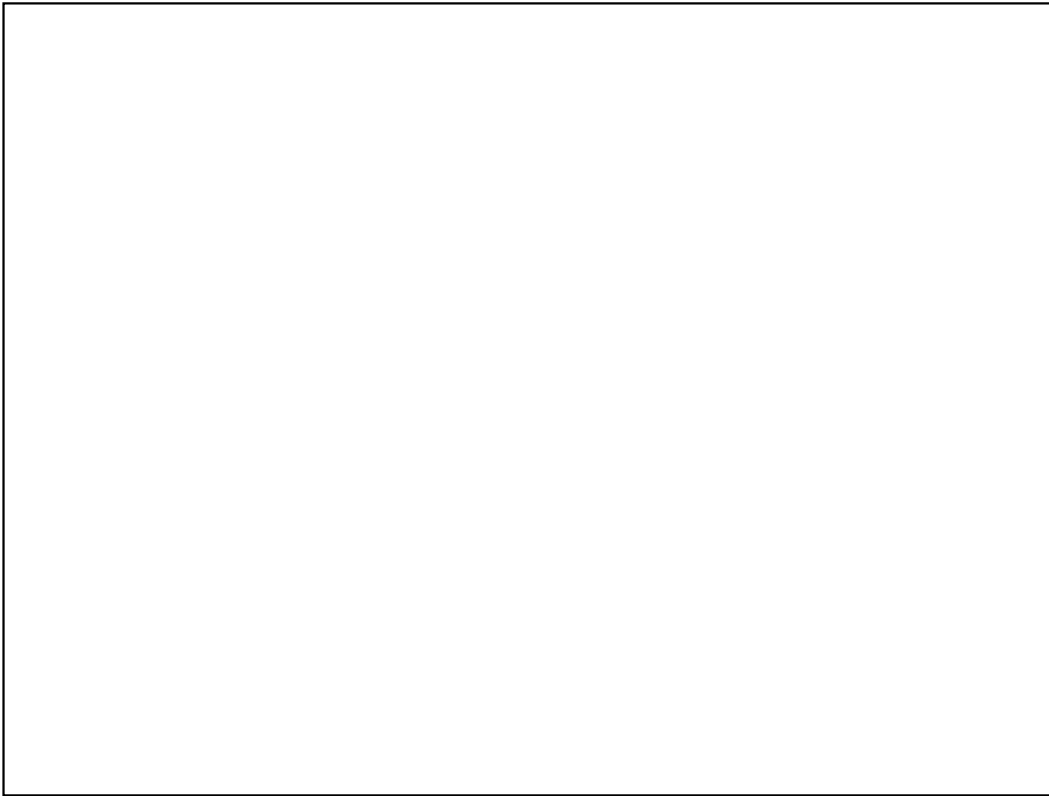
■ Diagnosis and Treatment

- Tissue biopsy culture, Blood culture-positive
- Blood tests-Leukocytosis, CRP elevated, Serum Creatine Kinase and AST elevated
- Imaging and blood work up
- Treatment: Antibiotics (Clindamycin+ Vancomycin+ Carbapenem) broad spectrum to cover Strep, Staph, GNB and Anaerobes) + Surgical debridement



spindle shaped
claustrophobic in the presence of oxygen
do not have catalase or superoxide dismutase - make oxygen toxic

GAS GANGRENE/ CLOSTRIDIAL MYONECROSIS



no spores in tissue sample bc in tissue does not require spores
some will look gram negative but are gram positive

C. tetanus is drumstick spore *C. perfringens* is subterminal spore



Lawn mover
injury to left foot



Right arm with open
wound-
Combat related
injury

High risk settings for Gas Gangrene

Posttraumatic exposure to soil

- Open wounds or fractures
- Gunshot wounds with foreign bodies
- Burns and frostbites
- IV drug abuse

Postoperative

- Bowel resection
- Biliary surgery

Spontaneous

- Colon Cancer(*C. septicum*)
- Neutropenia

PATHOGENESIS OF GAS GANGRENE

Contamination
of deep
wounds



Germination of
bacterial
spores



Conversion to
vegetative
forms



fermentation of
carbohydrates in
tissues

Abundant **gas**
formation



production of
necrotizing and
hemolytic exotoxins

12 toxins.
Most imp. **Alpha**
toxin also called
Lecithinase or
Phospholipase C



production of
extracellular
enzymes:

Collagenase
and
Hyaluronidase



anaerobic condition

1-3 days after injury

- ✓ **Acute and increasing pain**
- ✓ **Edema, gas** (crepitus in tissue)
- ✓ **Discoloration and hemorrhagic bullae**
- ✓ Systemic signs include fever, tachycardia, hypotension, shock
- ✓ Rapidly progressive and can be fatal

Alpha toxin leads to lysis of RBCs, WBCs, muscle cells, etc. There is increased vascular permeability, systemic absorption of toxins
→ hypotension → shock



GPC
GNB
aerobic -> used
up O₂

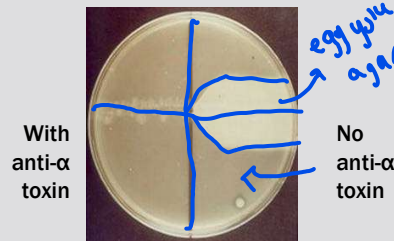


GAS GANGRENE- DIAGNOSIS

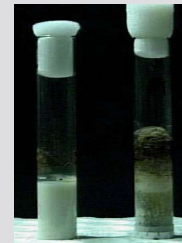
- **Clinical diagnosis**- discoloration, bullae, dishwater pus discharge, crepitus
- **Radiograph**-linear streaks of gas in soft tissue/ bubbles in tissue
- **Grams stain**-Wound swab shows GPB with box car morphology. Paucity of neutrophils, spores absent.
- **Anaerobic culture**- target beta hemolysis or double zone beta hemolysis.
- **Identification tests**



Blood agar
double zone
of hemolysis



Nagler reaction to
demonstrate *lecithinase*
(see notes section for details)



Stormy fermentation
in milk medium

Nagler's reaction or test is used to detect the alpha toxin responsible for gas gangrene. In Nagler's test, one half of the Egg Yolk agar is plate coated with antitoxin while the other half isn't. On that half of the plate where there is no antitoxin, opaque areas are seen around the colonies due to the action of toxin lecithinase (Phospholipase C) elaborated by *Cl.perfringens* on the lecithin present in Egg Yolk medium.



A CASE OF TETANUS



- A 12-year-old boy presents with stiffness of the jaw and neck along with inability to swallow.
- Twelve days ago he stepped on a rusty nail. This led to a punctured wound and the area is now swollen, red and painful. He has been experiencing tingling numbness and spasms in his calf muscles.
- O/E he has trismus (jaw rigidity), facial muscle spasms, dysphagia and neck rigidity. His back is arched producing opisthotonus.
- Labs: CBC-leukocytosis, CSF, Blood chemistries-normal
- **Diagnosis: Tetanus (generalized)**

- What is tetanus?
- What is the causal organism and its features?
- What is the virulence factor and its mode of action?
- How is the disease diagnosed, treated and prevented?

C.tetani- CHARACTERISTICS

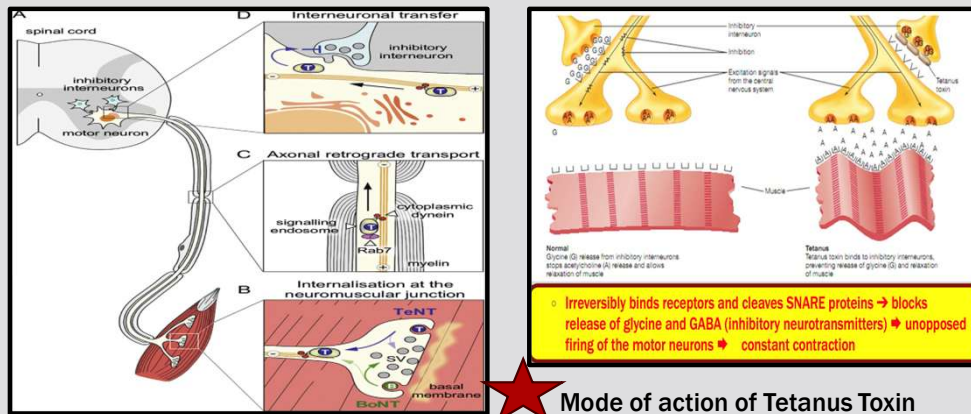


Gram stain of culture smear of *C.tetani*. Note the GPB with terminal spores giving them a tennis racket/ drum-stick appearance.

- GBP with **terminal spores** (tennis racket/ drum-stick appearance)
- **Obligate anaerobe.**
- Found in animal gut and contaminates soil/ dust (spores viable for many years)
- Transmission- puncture wounds/ trauma, needs low tissue oxygenation
- Elaborates important **Toxins**:
 - **Tetanospasmin/Tetanus Neurotoxin(TeNT)**-an AB exotoxin, high potency with lethal dose 2.5ng/kg.
 - **Tetanolysin** brings out lysis of other cells

TETANUS-PATHOGENESIS

Anaerobic conditions → spore germination → local bacterial multiplication
 → toxin production → toxin enters nervous system peripherally and is
 carried via retrograde **axonal transport** to **CNS**



Remember

TETANUS- CLINICAL FEATURES, DIAGNOSIS AND PREVENTION

Wound

No or minimal inflammation at the site of infection

Clinical findings

Spastic paralysis = strong muscle spasms

Early symptoms

Lockjaw (trismus) = spasms of masseter muscles

Risus sardonicus = grimace

Difficulty swallowing = pharyngeal muscle spasms

Late findings

Muscle rigidity

Opisthotonus = arching of back

Others- Generalized tetanus, Neonatal tetanus

Risus sardonicus



Lock jaw



Opisthotonus

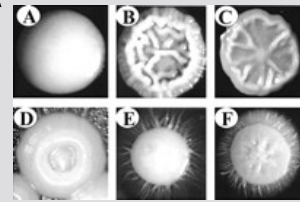


Treatment- wound treatment, antibiotics, muscle relaxants

Prevention- passive(tetanus immunoglobulins) and active immunization (Tetanus Toxoid).

FUNGAL MYOSITIS PATHOGENS

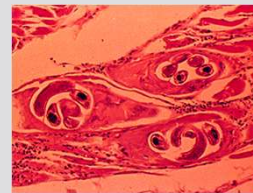
- **Candida** species
 - Most common cause
 - Budding yeast with blastoconidia
 - Smooth, white, **creamy colonies**
 - May undergo **phenotype switching to fuzzy colonies** and hyphae/pseudohyphae
- Increased risk due to
 - ↓ granulocytes
 - Chemotherapy
 - ↑ broad-spectrum antibiotics
- Often seen in conjunction with chronic mucocutaneous candidiasis
- Treatment = amphotericin B or azole



PARASITIC MYOSITIS
**Trichinellosis/
Trichinosis
and Cysticercosis**

Trichinella spiralis – PORK WORM

- Nematode parasite which cause Trichinellosis/ Trichinosis.
- In the US, ~1.5 million persons carry live *Trichinella* in musculature
- Risk factors: **Consumption of raw or undercooked meat** which is usually not sourced from commercially farmed food animals.
- Typical sources: **undercooked pork/ wild game meat-bear, wild boar, wildcat, fox, wolf, seal or walrus.**
- Homemade jerkies or sausages from such meat.
- **Infective form- Larvae encysted within muscles**



T. spiralis encysted in muscle
22

nematode parasite



Outbreak of Human Trichinellosis — Arizona, Minnesota, and South Dakota, 2022

Weekly / May 23, 2024 / 73(20):456–459

[Print](#)

Shama Cash-Goldwasser, MD¹; Dustin Ortahn, MPH²; Muthu Narayan, DO³; Conor Fitzgerald, MPH⁴; Keila Maldonado⁵; James Currie, MD⁶; Anne Straily, DVM⁷; Sarah Sapp, PhD⁷; Henry S. Bishop⁷; Billy Watson, PhD⁷; Margaret Neja⁷; Yvonne Qvarnstrom, PhD⁷; David M. Berman, DO⁸; Sarah Y. Park, MD⁸; Kirk Smith, DVM, PhD⁹; Stacy Holzbauer, DVM^{9,10} [\(VIEW AUTHOR AFFILIATIONS\)](#)

[View suggested citation](#)

Summary

What is already known about this topic?

Human trichinellosis cases in the United States are rare and are usually acquired through consumption of wild game.

What is added by this report?

Among eight persons who shared a meal that included the meat of a black bear harvested in Canada and frozen for 45 days, six trichinellosis cases were identified. The meat was grilled with vegetables and served rare; two cases occurred in persons who ate only the vegetables. Motile freeze-resistant *Trichinella nativa* larvae were identified in remaining meat frozen for >15 weeks.

What are the implications for public health practice?

Cooking meat to an internal temperature of $\geq 165^{\circ}\text{F}$ ($\geq 74^{\circ}\text{C}$) is necessary to kill *Trichinella* spp. parasites. *Trichinella*-infected meat can cross-contaminate other foods, and raw meat should be kept and prepared separate from other foods to prevent cross-contamination.

Article Metrics

Altmetric:



Field Citation Ratio	n/a
Relative Citation Ratio	n/a

[Figure](#)

FYI

Morbidity and Mortality Weekly Report

Two Outbreaks of Trichinellosis Linked to Consumption of Walrus Meat — Alaska, 2016–2017

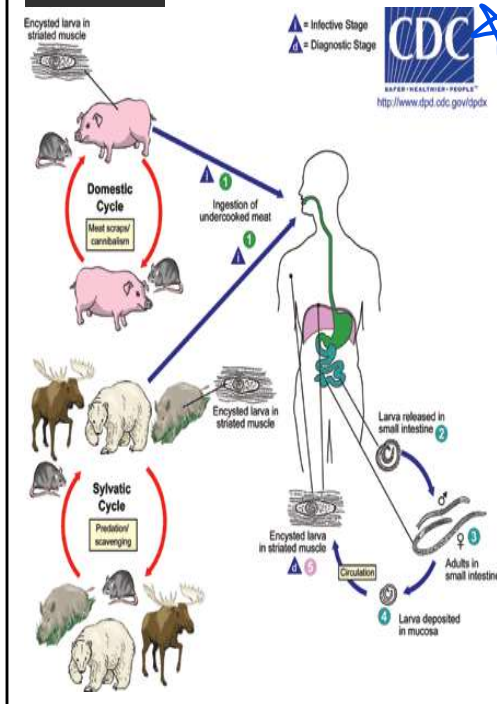
Yuri P. Springer, PhD^{1,2}; Shannon Casillas, MPH³; Kathryn Helfrich, MSN, MPH¹; Deanna Mocan⁴; Marscleite Smith⁴; Gabriela Arriaga⁴; Lyndsey Mixson^{1,5}; Louisa Castrodale, DVM¹; Joseph McLaughlin, MD¹

During 1975–2012, CDC surveillance identified 1,680 trichinellosis cases in the United States with implicated food items; among these cases, 1,219 were attributed to consumption of raw or pork products, and 461 were attributed to nonpork products.

On September 19, staff members of Norton Sound Regional Hospital in Nome reported two additional suspected trichinellosis cases from the same community, in the adult aunt (patient D) and uncle (patient E) of patient A. Both patients

Reference: MMWR / July 7, 2017 / Vol. 66 / No. 26

Life cycle and Pathogenesis



MOI: Ingestion of meat containing larvae

Mild to severe infection based on load of larvae and immune status of patient.

1-2 days- **Enteral phase.**

Release of larvae from meat > penetrate intestinal mucosa and mature into adult worms > copulation and release of larvae.

2-8 days later- **Parenteral phase**

Larvae migrate via blood stream to skeletal muscles.

Enteral phase: nausea, abdominal pain, diarrhea.

Parenteral phase:

- **Muscle pain**, swelling, weakness.
- Subungual **splinter hemorrhages**.
- **Eosinophilia**
- **More severe cases- myocarditis, encephalitis, thrombo-embolic phenomenon.**
- Immune response leads to adult worm expulsion, but larvae can persist for months-years.

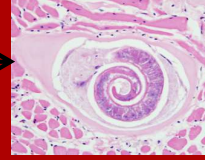
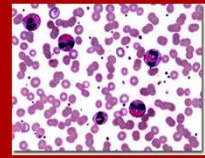
not required to know all the details of the life cycle

TRICHINOSIS

➤ Diagnosis



- Recent history of eating raw or undercooked meat
- Clinical presentation- including eosinophilia
- ✦ Detection of **specific antibodies in the serum**
- **Muscle biopsy**(infrequently done) for histopath microscopy to visualize encysted larvae. Also helps in molecular detection of species.



muscle biopsy
not always
possible

➤ Treatment



- Corticosteroid for treatment of inflammatory symptoms in severe disease
- **Albendazole** or Mebendazole

➤ Prevention



- Adequate cooking of meat

TAENIA SOLIUM – PORK TAPEWORM

- *T. solium* is a cestode- Tape worm with a scolex (head), flat body, segmented proglottids.
- Adult tape worm infestation is **Taeniasis** (acquired by eating measly pork)

■ Skin and soft tissue manifestations are seen due to **Cysticercosis** caused by the larvae of tapeworms (and not by the adult worms).

■ Mode of infection: **Ingestion of tapeworm eggs shed by another human being suffering from Taeniasis.**

■ **Eggs hatch in stomach** and release oncosphere. These penetrate intestinal wall → circulation → travel to tissue(muscles, CNS, eyes) and develop into **cysticerci**.



Scolex and body of adult tapeworm



Note the cysticerci over the trunk and shoulders

CYSTICERCOSIS

- **Clinical picture**

- Cysticerci in muscle: asymptomatic or nodules, inflammation
- Cysticerci in the eyes: loss of visual acuity
- Neurocysticercosis: seizures, and headaches

- **Diagnosis**

- Radiology
- Muscle biopsy- histopathology
- Antibody detection



- **Treatment**

- **Albendazole or praziquantel**
- Steroids to decrease inflammation
- Surgical removal of cysts



Muscle biopsy-
histopathology section.
Microscopy shows encysted
larvae

REFERENCES

- **Murray's Medical Microbiology 9th edition**
 - *Staphylococcus aureus* – Ch.18
 - *Clostridium* spp – Ch.30
 - *Neisseria gonorrhoeae* – Ch.23
 - *Borrelia burgdorferi* – Ch.32
 - *Candida* spp. – Ch.65
 - *Trichinella spiralis* – Ch.75
 - *Taenia solium* – Ch.77
- CDC website for parasites
- www.uptodate.com

